AI IN SPACE AND ORBITAL OPERATIONS - VOLUME 7

The AI Lunar and Deep-Space Operations Playbook

How AI operates where help is minutes to hours away - lunar surface, cislunar space and deep space - without ever mistaking a simulation for survival.



Gates LD0-LD5 | Eight templates | 20 golden rules

A simulated landing is not a landed mission. Lunar and deep-space AI is licensed per destination, per vehicle, per light-time regime and per mission phase - never in general, and never on an Earth-analogue test alone.

The claim ladder for this volume: planned, simulated, rehearsed, survived, transferable. A system earns the rung it can prove at real distance, in a real environment, on a real clock - not the rung its simulation reached.

Beyond Earth orbit, the ground stops being a safety net and becomes a correspondent; the question is what the vehicle may decide alone

Every earlier volume assumed help was seconds away. Lunar operations break that gently - about 1.3 seconds each way, blackouts behind the Moon, a surface no one can walk to. Deep space breaks it completely: minutes to hours of light-time, contacts days apart, a vehicle that must survive its worst hour entirely alone. AI is not optional out there; unbounded, unlicensed AI is.

- Name the destinations, vehicles, light-time regimes and phases in scope - a licence for lunar orbit says nothing about the surface, and nothing about Mars cruise.
- State the claim on the ladder: planned, simulated, rehearsed, survived or transferable - and the evidence that would move it.
- List the standing refusals on page one: no unbounded trajectory or landing-site change, no learning in flight during descent or other critical phases, no planetary-protection shortcut.
- Price the downside in missions lost, not in contact hours: at deep-space distance there is no rescue, no patch truck and no second vehicle.

This volume licenses distance autonomy the way Volume 1 licensed space AI and Volume 6 licensed the ground: per destination, per vehicle, per light-time regime and per phase, with envelopes signed before departure and demotion built in. Distance does not lower the standard. It raises it, because distance removes the human who used to hide the gaps.

Rule: if a decision cannot wait for Earth, it must be decided inside an envelope Earth signed in advance - or it must not be taken at all.

EXECUTIVE LENS

Distance is a design input, not an operations inconvenience; budget autonomy like propellant.

MISSION LENS

A licence names a destination and a phase; 'deep-space ready' names nothing.

OPERATIONS LENS

Crews and controllers trust what survived, not what simulated beautifully.

Ten stages from translunar injection to end of mission, graded stage by stage

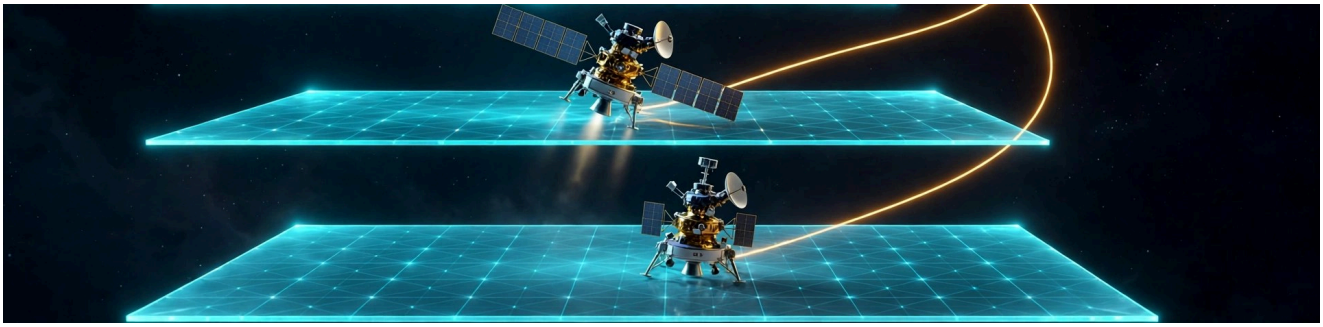


Figure 1 - One vehicle in five states - cruise, approach, descent, surface and long cruise - each on its own time plane, linked by an amber trajectory thread.

Grade each stage separately. A cruise autonomy that survived six quiet months has earned nothing about descent, and a brilliant landing system has earned nothing about surviving a lunar night.

- 1. Departure and translunar / interplanetary injection. 2. Cruise navigation and health. 3. Approach and orbit insertion. 4. Descent, landing and ascent. 5. Surface mobility and operations.
- 6. Power, thermal and night / eclipse survival. 7. Communications, relays and delay-tolerant networking. 8. Science, sampling and resource operations. 9. Rendezvous, docking and cislunar traffic. 10. End of mission, disposal and planetary-protection close-out.

Stages 4 and 6 carry irreversible, time-critical risk and stay under the tightest licence in this volume. Stages 2, 7 and 8 are where AI earns its keep first: patient pattern work across long, quiet clocks, with Earth reviewing at leisure what the vehicle did alone.

EXECUTIVE LENS

Map by phase before you automate; most deep-space losses are phase-transition losses.

MISSION LENS

Each stage inherits the environment of its destination, not of the test range.

OPERATIONS LENS

The map is per vehicle; a fleet claim across destinations is a claim about none of them.

The speed of light is the senior operations manager on every deep-space mission, and it does not negotiate

One-way light-time is about 1.3 seconds to the Moon, roughly 3 to 22 minutes to Mars depending on geometry, and hours to the outer system. Every operational concept is a bet placed against that delay: what Earth sees is the past, what Earth sends arrives in a different present, and during solar conjunction or lunar farside passes there is no bet at all - only silence, planned or unplanned.

- Every procedure, alarm and autonomy claim states the light-time regime it was licensed in; a manoeuvre licensed at lunar delay is not licensed at Mars delay.
- Consoles display state age as first-class data: 'this telemetry is 18 minutes old' precedes every decision, never follows it.
- Blackout and conjunction plans are pre-signed missions in their own right, with entry conditions, permitted actions and exit checks - not gaps in the plan.
- Store-and-forward is the default mental model: commands are letters to a future vehicle, written, checked and dated as such (Volume 6).

Teams that internalise delay design calm missions. Teams that fight it design heroic ones, dependent on a contact, a geometry or a person being awake - and deep space eventually collects on every hero dependency.

Rule: design for the worst light-time and the longest silence in the mission, not the average contact. Averages do not attend anomalies.

EXECUTIVE LENS

Put light-time on the board pack; executives fund what they can picture arriving too late.

MISSION LENS

State age on telemetry is honesty infrastructure, like covariance in Volume 5.

OPERATIONS LENS

A planned silence is an operation. An unplanned one is an incident.

Autonomy is allocated by clock and consequence, tier by tier, with automatic demotion when the world stops matching the model

Distance forces autonomy, but it does not force any particular amount. The allocation question is precise: which functions must complete before Earth could possibly reply, which may proceed unless Earth objects, and which must always wait - even if waiting costs the opportunity. Volume 2 set function allocation in design; this chapter licenses it in operations, tier by tier.

- Tier the functions in writing: T1 advise Earth, T2 act with Earth approval inside the contact, T3 act inside a signed envelope and report, T4 act for survival only, deterministic and bounded.
- Every T3 envelope carries hard limits, an expiry and abort conditions; expired envelopes revert the function to a safer tier automatically, never by meeting.
- Demotion is automatic on triggers - sensor disagreement, environment outside the rehearsed range, model confidence collapse - and promotion back requires Earth review and evidence.
- Learning in flight is advisory only outside survival functions, and frozen entirely during critical phases: descent, docking, entry and night entry.

The honest test of a tier is its demotion behaviour. Autonomy that can only go up in authority is not a tier system; it is a ratchet pointed at a spacecraft no one can reach.

EXECUTIVE LENS

Fund demotion engineering like autonomy engineering; it is half the product.

MISSION LENS

Envelopes expire. Authority that never expires will eventually be used outside its evidence.

OPERATIONS LENS

Freeze learning in critical phases; novelty belongs in rehearsal, not in descent.

There is no positioning constellation waiting at the Moon or Mars; the vehicle navigates by what it can see, and proves it

Beyond Earth orbit, navigation becomes an onboard craft again: optical navigation against stars, planets and moons, terrain-relative navigation over mapped and unmapped ground, radiometric tracking when a link exists, and dead reckoning that drifts honestly in between. AI fuses these sources well - and can also fuse them into a confident, consistent, wrong position if one source is silently biased.

- Every navigation solution ships with its uncertainty and source health; consumers - guidance, science, comms pointing - must refuse solutions that arrive without them.
- Sources are cross-checked, not just averaged: disagreement between optical, radiometric and inertial solutions is surfaced as a navigation event, not smoothed away.
- Terrain-relative navigation is licensed per terrain class and illumination regime; a system proven over sunlit maria has earned nothing about a shadowed polar crater.
- Maps are versioned mission data with provenance (Volumes 4 and 6): which orbiter, which season of light, which processing - because the vehicle will bet its descent on them.

Navigation at distance is an evidence discipline. The vehicle that knows how lost it might be can slow down, safe itself or ask for a fix. The vehicle that is precisely wrong cannot.

EXECUTIVE LENS

Buy navigation on scored accuracy against withheld ground truth, per terrain and light.

MISSION LENS

Source disagreement is a product; surface it before guidance consumes it.

OPERATIONS LENS

Map provenance is flight safety data at descent, not metadata.

The minutes no one can help with: licensed like a launch, rehearsed like a landing, never improvised by a model

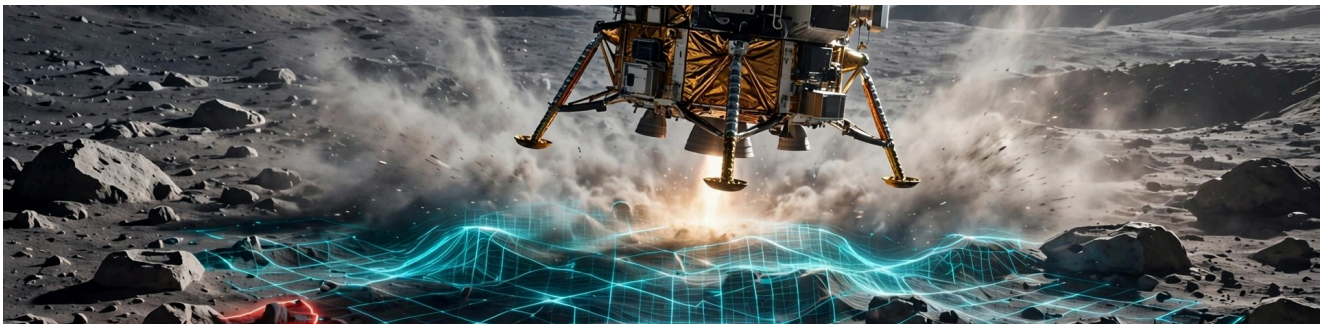


Figure 2 - A lander in final descent over its cyan terrain grid, amber landing ellipse beside the red hazard zone.

Descent is the purest case in this series: a short, irreversible, high-energy phase in which Earth is a spectator with delayed vision. AI earns its place here - hazard detection, terrain-relative navigation, divert decisions inside a signed ellipse - under the strictest licence in the volume, because there is no abort to a safe orbit once the ground is close and the propellant is spent.

- The landing envelope is signed before descent: ellipse, hazard thresholds, fuel reserves, divert limits and the conditions for a wave-off while one still exists.
- Hazard models are validated per terrain class, illumination and dust regime, with failure-seeking test campaigns (Volume 2), not demonstration landings on friendly ground.
- Onboard decisions in descent are deterministic where they are survival-critical; learned components advise within bounded authority and their inputs are monitored for out-of-distribution terrain.
- Ascent and rendezvous from the surface are licensed separately from landing: arriving intact proves nothing about leaving.

Every successful landing raises the temptation to widen the envelope on confidence alone. Resist it in the register: the envelope widens on evidence - dispersed simulation, analogue trials, as-flown reconstruction - or it does not widen.

Rule: in descent, AI chooses only among futures Earth already accepted. A landing site invented in the last ninety seconds is not a decision; it is a gamble with a mission as the stake.

EXECUTIVE LENS

Judge landing AI on its worst dispersed case, never its nominal run.

MISSION LENS

Divert authority is a signed, bounded gift; write its edges before flight.

OPERATIONS LENS

Reconstruct every descent as-flown; the next envelope is built from that data.

A rover's world is the few metres it can see; AI plans traverses the way a mountaineer plans a route - conservatively, and in daylight

Surface mobility trades Earth's certainty for local knowledge: wheel slip in regolith, rocks the orbiter never resolved, slopes that change with light, dust on every surface. AI path planning, visual odometry and hazard avoidance are genuinely enabling here, and their characteristic failure is optimism - a traverse that is feasible on the map and impossible at the third wheel.

- Traverses are planned in verified segments: drive only as far as stereo vision and slip models have actually characterised, then re-plan from new ground truth.
- Slip, sinkage and power draw are learned per terrain and fed back into planning margins; a rover that does not update its own difficulty model is repeating its first day forever.
- Every drive plan carries a retreat or safe-park option that survives the failure it is retreating from - comms loss, nightfall, a stuck wheel.
- Human approval tiers scale with commitment: a repositioning hop, a one-way descent into a crater and a night drive are three different licences, not one driving licence.

Distance travelled is a vanity metric on another world. Verified metres - driven, characterised and survivable in reverse or in place - are the currency surface operations should count.

EXECUTIVE LENS

Count verified metres, not odometer metres, in every report upward.

MISSION LENS

Retreat options are part of the plan, priced in energy, not an afterthought.

OPERATIONS LENS

Re-plan from ground truth at every segment boundary; maps age in metres out there.

The lunar night is fourteen Earth days of cold and dark that no command can shorten; survival is a budget signed in advance

At the lunar equator, night lasts about fourteen Earth days and temperatures fall far below what most electronics forgive. In deep space, sunlight itself thins to a suggestion and power becomes a permanent negotiation. AI manages these budgets well - load forecasting, hibernation sequencing, heater prioritisation - and fails dangerously when it optimises a budget it was allowed to redefine.

- Survival budgets are signed, central and unspendable by optimisers: reserve energy, heater minimums and wake-up margins are constraints, not variables (Volume 2).
- Hibernation and wake-up are rehearsed end to end, including the failure that matters: the vehicle that does not wake, and how Earth would ever know why.
- AI load forecasts are graded against as-flown consumption per season and terrain; chronic optimism demotes the forecaster to advisory.
- Thermal models are validated against the coldest and hottest cases the destination offers, not the cases the test chamber found convenient.

Most surface missions do not die dramatically. They run out of energy, in the dark, slightly earlier than a spreadsheet promised. This chapter exists to make that spreadsheet honest while Earth can still change it.

Rule: no optimiser spends the survival reserve. A plan that needs the reserve to work is not a plan; it is the incident report, written early.

EXECUTIVE LENS

Price night survival visibly; averaged into operations, it will be traded away.

MISSION LENS

Grade forecasts as-flown, per season; optimism is a measurable defect.

OPERATIONS LENS

Rehearse the no-wake case on the ground, where it is still hypothetical.

At distance, the network is intermittent by physics; design for messages that arrive late, out of order, or after the crisis they describe

Deep-space communications run on scarce, scheduled resources: Deep Space Network apertures shared across missions, lunar relays still being built, occultations, conjunctions and weather on Earth. Delay-tolerant networking turns that reality into a system - bundles stored, forwarded and prioritised across hops. AI scheduling and triage help, under discipline: the most important message in the queue is often the anomaly report, not the prettiest science product.

- Priority classes are pre-signed and survival-first: health and safety telemetry outranks science, science outranks housekeeping bulk, and the classes are tested under congestion, not just listed.
- Contact plans include the missed-contact case as a designed state: what the vehicle does, stores and drops when the pass does not happen.
- Data triage onboard is licensed with audited discard samples (Volume 4): what the vehicle chose not to send is reviewable, because at distance a discard can be a deletion.
- Relay dependence is mapped like any single point of failure (Volume 6): whose relay, whose priority rules, what happens when it serves someone else first.

Bandwidth at distance is a mission resource like propellant: finite, allocated and gone once spent. Treat every downlink plan as a budget with an owner, and every silence as a scheduled event with a plan of its own.

EXECUTIVE LENS

Test priority under congestion; a queue policy untested at overload is a wish.

MISSION LENS

Missed contacts are a design case, rehearsed, not a surprise.

OPERATIONS LENS

Audit onboard discards; at distance, triage is editing the mission record.

Science at distance is a chain of custody across millions of kilometres; AI may choose targets inside a signed intent, never rewrite the intent

Autonomous science - target selection, instrument scheduling, event detection, sample triage - multiplies the return of missions Earth can only steer in retrospect. It also creates the volume's subtlest risk: a vehicle that diligently collects what its model finds interesting, while the science the mission was funded to do quietly goes uncollected.

- Science intent is signed as a charter: questions, priorities, quotas for serendipity versus directed targets, and the trade rules when they conflict.
- Target selection is audited against the charter on a cadence: what was chosen, what was passed over, and whether the model's taste is drifting from the mission's questions.
- Sample selection, caching and return decisions carry full provenance - context imagery, location, instrument record - because a sample without context is a rock.
- Event-driven replanning (a plume, a quake, a fresh crater) operates inside pre-signed opportunity envelopes with energy and risk budgets attached.

The measure is not observations taken but questions answered, with the chain of custody intact. AI that returns more data and less science has optimised the wrong mission, efficiently, at great distance.

EXECUTIVE LENS

Audit the model's taste against the charter; drift is visible in the pass-over list.

MISSION LENS

Provenance makes a sample science; without it, mass is just mass returned.

OPERATIONS LENS

Opportunity envelopes let the vehicle be brave inside a budget Earth set.

Making oxygen, water or propellant from regolith is a chemical plant run by robots on another world; license it as a plant, not a demo

ISRU promises to break the tyranny of launched mass: water ice from shadowed craters, oxygen from regolith, propellant made where it will be burned. AI controls excavation, processing and quality well in principle. The licensing trap is scale theatre - a gram produced in a demonstration reported as a capability, and a crewed return plan quietly built on a plant that has never run a full season.

- ISRU claims are stated as sustained rates with uptime, feedstock grade and product quality - verified over a full environmental cycle, not peak hourly output.
- Excavation and feedstock AI is licensed per terrain and ice / regolith model; a resource map is a hypothesis until a vehicle has dug where the map said to dig.
- Product quality is measured, not inferred: propellant or life-support feedstock that will carry crew is certified by instrument, with off-spec product isolated automatically.
- No crew or return plan may depend on ISRU output beyond the rung actually survived; aspirational production is planned as a bonus, never as the baseline.

ISRU will change everything when it works at scale, which is exactly why its claims must be kept boringly honest now: missions planned on rung-2 production with rung-4 confidence are how programmes strand people and vehicles far from home.

Rule: depend only on what the plant has already produced, sustained, at grade. Hope is not a feedstock.

EXECUTIVE LENS

State sustained rate, uptime and grade together; any one alone is marketing.

MISSION LENS

Dig where the map says before you license the map.

OPERATIONS LENS

Certify crew-bound product by instrument, every batch, automatically.

When the passenger is a person, the autonomy argument changes its moral weight: AI advises, protects and never quietly trades a crew margin

Crewed lunar and deep-space missions put AI inside the life-support loop: atmosphere and water monitoring, radiation shelter management, medical support, habitat fault detection during the long hours a small crew must sleep. The technology overlaps the robotic case; the licence does not. A system that may inconvenience a rover may not gamble with a crew's air.

- Life-support AI is licensed to detect, alarm and act protectively inside deterministic bounds; optimisation of consumables never overrides signed crew margins.
- Crew override is absolute, immediate and physical: any habitat automation can be isolated by hand, and the isolated state is a designed, livable state.
- Medical and behavioural support tools are advisory to the crew, with their limits stated plainly; no diagnosis or treatment is automated beyond pre-signed emergency protocols.
- Radiation and shelter decisions use conservative, pre-agreed triggers; a model may recommend sheltering earlier, never later, than the signed threshold.

The crew is the ultimate local autonomy: trained, present and accountable. Habitat AI earns its place by making that crew harder to surprise - earlier warnings, cleaner state, quieter nights - and by knowing, in architecture, that it is not in command of the air they breathe.

EXECUTIVE LENS

Crew margins are signed constraints, not optimisation variables; audit any system that touches them.

MISSION LENS

Design the isolated habitat first; automation is added to a place people can already survive.

OPERATIONS LENS

Earlier shelter is a recommendation AI may make. Later is not.

Cislunar space is becoming a neighbourhood: gateways, landers, tugs and stations sharing orbits with no tower to call

As lunar activity grows, vehicles must find, approach and dock with each other around a body with lumpy gravity, sparse navigation aids and partners built by different nations and companies. AI relative navigation and trajectory planning are enabling - and a misjudged approach at distance is a collision that creates the debris problem of Volume 5 in the worst possible neighbourhood.

- Approach corridors, hold points and abort trajectories are agreed and published between operators before flight; AI plans inside the corridor, never negotiates a new one in flight.
- Relative navigation is licensed per target class - cooperative with fiducials and transponders is a different licence from uncooperative capture of a silent object.
- Every approach carries an abort that is safe under total AI failure at the worst moment; passively safe trajectories are the default, powered safety is the exception.
- Cislunar coordination follows Volume 5 discipline: commitment locks, shared ephemerides and human-decided collision manoeuvres, extended to surface and orbital assets alike.

The cislunar commons will be built by whoever behaves predictably in it. Licensing approach AI on passive safety and published corridors is how an operator earns the right to be approached by everyone else.

EXECUTIVE LENS

Passive safety first: an abort that needs the AI to work is not an abort.

MISSION LENS

Publish corridors and hold points; predictability is a public good in cislunar space.

OPERATIONS LENS

Cooperative and uncooperative targets are separate licences, full stop.

Some mistakes cannot be recalled because they propagate biologically and chemically; protection rules outrank mission appetite

Planetary protection is the discipline of not contaminating the worlds we study or the world we return to: forward contamination of Mars, Europa or Enceladus could destroy the science and possibly something living; back contamination is a public-safety duty on sample return. AI enters in sterilisation verification, cleanroom monitoring, sampling decisions and trajectory biasing - always as a servant of the protection plan, never as its editor.

- Protection categories and requirements are set by the mission's COSPAR-based plan and national authority; no onboard optimisation may trade a protection constraint for science, mass or schedule.
- Sterilisation and bioburden records are provenance data with the same status as flight data (Volume 6): complete, auditable and attached to the hardware's identity.
- Sampling AI avoids protected features - recurring slope lineae, plume sources, special regions - by signed exclusion zones, with violations treated as mission-level incidents.
- Sample-return containment is deterministic and independently verified; learned systems may monitor, never gate, the containment chain.

Protection failures are unique in this series: they cannot be patched, recalled or compensated, and they may not be discovered for decades. That asymmetry is why this chapter's rules are written as refusals, not guidelines.

Rule: when science appetite and planetary protection conflict, protection wins, automatically, without a meeting at light-time distance.

EXECUTIVE LENS

Protection constraints are hard-coded exclusions, not weighted preferences.

MISSION LENS

Bioburden provenance travels with the hardware record for the life of the mission.

OPERATIONS LENS

Containment is monitored by AI and gated by determinism; keep that order.

FDIR at distance must assume the fault, the silence and the wrong diagnosis can all happen in the same week

Fault detection, isolation and recovery in Earth orbit assumes a ground team in the loop within minutes. At distance, the vehicle must detect, decide and recover - or safe itself and wait, possibly for days, in a state that does not decay while it waits. AI diagnosis helps prioritise and pattern-match; survival logic stays deterministic, layered and deliberately simple.

- Survival FDIR is deterministic and independent of learned components, as in Volumes 1 and 3: it safes the vehicle on limits, not on interpretations.
- Safe states are designed for duration: a deep-space safe mode must be stable for the longest plausible silence, thermally and in power, not just for a weekend.
- AI diagnosis is graded on drills and as-flown cases with its uncertainty shown; a confident wrong isolation that powers down the wrong string is worse than no diagnosis.
- Recovery is staged and reversible where physics allows: Earth rehearses the recovery sequence against the twin before the vehicle is asked to perform it alone.

The design question for every fault is triple: can the vehicle survive it alone, can Earth understand it from the data that will eventually arrive, and can recovery be attempted without making a second fault fatal? License only where all three answers are evidence, not optimism.

EXECUTIVE LENS

Safe for the longest silence, not the average contact gap; size it in days and seasons.

MISSION LENS

Grade diagnosis in drills; in flight is too late to learn its error rate.

OPERATIONS LENS

Rehearse recovery on the twin before the vehicle performs it alone.

An update sent to a vehicle you cannot touch is a letter that rewrites its recipient; write it, check it and seal it accordingly

Deep-space commanding inherits Volume 6 whole - independent checking, paired versioning, rehearsed rollback - and adds distance: no quick patch, no watching it land in real time, and contacts too scarce to waste on an upload that has to be sent twice. AI assistants that generate sequences or patches face the strictest version of the ground rules.

- Command loads and updates are verified by the independent checker (Volume 6, Chapter 5) and flown first on the high-fidelity twin at the exact flight configuration, across a simulated full contact cycle.
- Uploads are staged, checksummed and activatable on a future command or condition, with a tested return to the prior image that does not itself require a healthy new image.
- Activation windows avoid critical phases and long silences: nothing new wakes up just before a blackout, a night or a descent.
- Every load states its light-time assumptions: what the vehicle should do if the confirmation contact is missed, and when it should revert on its own.

Distance turns release discipline from a virtue into a survival trait. The programmes that patch calmly at 20 light-minutes are the ones that rehearsed boredom at zero.

Rule: never activate what you cannot revert, never revert by a path you have not tested, and never do either on the edge of a silence.

EXECUTIVE LENS

Twin-first, at exact configuration, over a full simulated contact cycle; no exceptions for small patches.

MISSION LENS

Missed-confirmation behaviour is part of the load, signed with it.

OPERATIONS LENS

Schedule activations for watched time; novelty and blackouts do not mix.

Earth can imitate the Moon's light and Mars's delay, but not their indifference; rehearsal evidence is graded by what it cannot prove

Distance missions are rehearsed in an unmatched toolkit: high-fidelity simulators and digital twins (Volume 2), analogue sites in deserts, lava fields and polar stations, delay-injected operations, and hardware in thermal-vacuum and regolith bins. AI both uses these tools and is tested by them. The discipline is honesty about the gap: every analogue proves some things and is silent about others.

- Every rehearsal claim lists its fidelity gaps beside its result: gravity offloaded, dust simulated, delay injected, radiation absent - the gap list is part of the evidence.
- Operations rehearsals run at real light-time with real console discipline, including nights, handovers and the missed contact; a team that has only rehearsed at zero delay has not rehearsed.
- Failure-seeking campaigns target phase transitions and silences specifically: descent with a sensor dropout, night entry with a degraded battery, conjunction with a sick radio.
- Analogue success licenses the next rehearsal, not the flight claim: rungs are earned in order, planned, simulated, rehearsed, survived (Chapter 1).

The twin and the test range are where distance missions are allowed to fail. Programmes that protect their rehearsals from failure import that failure to a place where it costs a spacecraft.

EXECUTIVE LENS

Publish the fidelity-gap list with every rehearsal result; silence about gaps is a gap.

MISSION LENS

Rehearse at real delay with tired crews; that is the operational truth.

OPERATIONS LENS

Analogue success buys a harder rehearsal, never a flight rung by itself.

The team is split by physics: intent travels out, judgement is exercised locally, and trust is built from reported outcomes

Human-robot teaming at distance is not teleoperation. It is supervision across a delay that turns every exchange into correspondence: Earth sends intent, constraints and questions; the vehicle sends state, choices made and evidence. Designing that correspondence - what the vehicle reports unasked, what it must ask about, what it must never do unreported - is the real interface work of deep-space AI.

- The reporting contract is explicit: decisions above a signed significance threshold are reported with their basis at the next contact, whether they succeeded or not.
- Intent is sent as goals with constraints and margins, not as joystick sequences; the vehicle's plan to meet the intent is returned and reviewable before execution where the clock allows.
- Trust is calibrated from the as-flown record (Chapter 21's accounting): functions with accurate self-reports earn wider envelopes; inaccurate reporters are demoted, however capable.
- Crew and ground roles are rehearsed for the disagreement case: Earth believes the vehicle is wrong, the vehicle believes its sensors, and the protocol - not seniority at a distance - decides the next step.

Good correspondence makes a distant vehicle feel like a trusted colleague: predictable, candid about uncertainty, prompt to confess. Bad correspondence makes even a capable vehicle a liability its own team starts to route around.

EXECUTIVE LENS

Write the reporting contract before flight; significance thresholds are not improvised mid-mission.

MISSION LENS

Self-report accuracy is the trust metric; grade it like navigation.

OPERATIONS LENS

Rehearse the disagreement protocol; at 20 minutes delay, protocol is the relationship.

Simulations are polite; environments are not: radiation flips bits, dust abrades seals and cold rewrites material properties

Deep-space and lunar environments attack the assumptions AI is built on. Galactic cosmic rays and solar particle events upset electronics and sensors. Lunar dust - sharp, electrostatic, pervasive - degrades mechanisms, optics and radiators. Temperature extremes cycle every joint and battery. A model validated in a clean lab has met none of its real examiners.

- Compute and model integrity are monitored in flight: error-correcting memory, watchdogs and plausibility checks on sensor and model outputs, with radiation events logged as context for every anomaly.
- Dust mitigation is an operations discipline with AI in support: exposure budgets for mechanisms and optics, cleaning or avoidance behaviours, and degradation tracked as a consumable.
- Environmental envelopes for each AI function state the tested ranges - temperature, radiation, dust, illumination - and the function's behaviour outside them is demotion, not extrapolation.
- Solar storm procedures are pre-signed for crew and vehicle alike: shelter, safe configurations and the data that may be distrusted afterwards.

Environment is the reason the ladder's fourth rung is called survived. Until a system has operated through the destination's real weather - its storms, nights and dust - its claim, however brilliant, remains a rehearsal claim.

EXECUTIVE LENS

Log radiation context with every anomaly; some faults are weather reports.

MISSION LENS

Track dust degradation as a consumable with a budget, not as wear to discover.

OPERATIONS LENS

Out-of-envelope means demote; extrapolation at distance is fiction with authority.

No one owns the Moon, everyone is responsible for their spacecraft, and partners multiply both capability and veto points

Lunar and deep-space operations sit inside a real legal and political frame: the Outer Space Treaty makes states responsible for their national activities, including commercial ones; the Artemis Accords and national licensing shape conduct, heritage protection and resource activity among signatories; spectrum and DSN access are allocated, negotiated resources. International and commercial partnerships add interfaces no AI may paper over.

- Mission authorisation names the responsible state, licence and protection category before flight design closes; legal review is a gate input (LD0), not a launch-week errand.
- Resource, heritage and safety-zone claims are stated modestly and precisely, in line with the operator's actual agreements; AI-generated filings and reports receive the same legal review as human drafts.
- Partner interfaces - navigation standards, docking, comms, data rights - are contracted with the same discipline as Volume 6 vendor terms, plus the sovereign case: what happens if a partner withdraws.
- Spectrum, relay and DSN allocations are treated as mission-critical supply chains, with contention and priority understood in writing before cruise, not during an anomaly.

Distance does not dilute responsibility; it concentrates it on whoever signed the authorisation. Programmes that treat law and partnership as paperwork discover, at the worst contact, that they were load-bearing.

EXECUTIVE LENS

Authorisation and protection category at LD0; lawyers in the design phase are cheaper than lawyers in the anomaly phase.

MISSION LENS

State resource claims as agreed, not as hoped; precision is diplomatic capital.

OPERATIONS LENS

Contract the partner-withdrawal case while relations are warm.

At distance the scarce currencies are contacts, energy and time itself; count autonomy in verified days survived and science returned

Count distance economics the series way, in three columns: savings claimed, savings verifiably released, and value banked. The unit is the verified mission day: a day in which the vehicle survived inside its budgets, executed its signed intent, and returned its record with provenance - divided into fully loaded cost, including rehearsal, DSN and relay time, twin upkeep and the ground team that distance was supposed to shrink but must honestly retain.

- Fully loaded cost includes contact time at its real scarcity price, rehearsal and twin maintenance, and the anomaly team retained for the silences - the costs autonomy proposals most often assume away.
- Savings from reduced ground staffing are claimed only at the tier actually licensed: a T3 function still supervised every contact has not released the supervisor.
- Science value is counted against the charter (Chapter 10): verified days that answered no signed question are survival days, reported as such, not science days.
- Longevity is banked value only when it is usable: extra months beyond design life count when budgets, degradation and contacts support real operations in them.

Done honestly, distance autonomy usually pays - contacts are scarce and vehicles are patient. Done on vendor arithmetic, it pays until the first long silence reprices the ground team that was released too early.

Rule: cost per verified mission day, with rehearsal, contacts and the retained anomaly team inside the number. A saving bought by thinning any of them is a loan from the silence budget.

EXECUTIVE LENS

Price DSN and relay time at scarcity, not at tariff; the queue is the cost.

MISSION LENS

Claim staffing savings at the tier survived, not the tier designed.

OPERATIONS LENS

Survival days and science days are different currencies; report both.

Anyone can halt the distance autonomy before departure; after departure, the signed envelope is the halt, and it must be built to be trusted

The operating model binds the volume together: a distance operations authority owning the licence per destination, vehicle and phase; envelopes and reporting contracts signed before departure; rehearsal manufacturing the evidence; and a record - the operations wall of Figure 3 - on which every function's rung, envelope and self-report accuracy are visible at a glance.

- Roles: distance operations authority, mission lead, autonomy envelope owner, planetary-protection officer and navigation owner - named people, with deputies, per mission.
- Every licence states its rung, envelope, expiry, demotion triggers and reporting contract; demotion is automatic on triggers, promotion is earned at a gate with evidence.
- Pre-departure halts are drilled and honoured without blame: stopping a launch or a descent for envelope doubt is a success state, and the record says so.
- In flight, the halt lives in the design: envelopes expire, tiers revert, safe states wait - because at distance, no human hand is on the switch, so the switch must be worthy of the hand.

Earth orbit let us supervise our machines. The Moon asks us to trust them a little, and deep space asks us to trust them for months at a time. This operating model exists so that trust is never a feeling: it is a register, an envelope and a record of promises kept at a distance no one can shout across.

EXECUTIVE LENS

Name the authority per mission; at distance, unclear ownership arrives too late by definition.

MISSION LENS

Automatic demotion, gated promotion - the ratchet only ever turns toward evidence.

OPERATIONS LENS

Honour pre-departure halts in the record, or the next one will not be called.

WORKED EXAMPLE - AURORA ORBITAL, PART 1

The audit: a lunar polar landing and a deep-space cruise meet their own envelopes



Figure 3 - The Aurora distance operations wall: lander, rover, habitat and probe, one probe cell held in red beside the light-time clock.

Fictional worked example. All organisations, figures and events are illustrative.

Aurora Orbital (Volumes 1-6) flew two distance missions on one autonomy platform: Selene, a lunar polar lander and rover targeting water ice, and Wayfarer, an outer-system cruise probe. The vendor claimed a 49 percent operations saving and 54,000 ground hours per year, with 'full autonomy from injection to surface science'.

What the audit found

- Verified saving was 17 percent: 23,800 hours verifiably released and 14,600 banked, once rehearsal, twin upkeep, DSN time and the retained anomaly team were costed back in.
- Selene's hazard model, validated on sunlit mid-latitude terrain, scored 31 percent worse in shadowed polar illumination in failure-seeking tests; descent was licensed only after a polar-specific campaign and a smaller signed ellipse (Chapter 6).
- Wayfarer's science target selection drifted from its charter: 42 percent of chosen targets sat outside the signed questions, visible only when the pass-over list was audited (Chapter 10).
- A hibernation rehearsal found the lander would not have woken after a degraded-battery lunar night; the survival reserve had been treated as an optimisation variable by the power planner (Chapter 8).

None of this required new physics. It required the envelope register to be believed, polar light to be rehearsed in, and a pass-over list to be read - the unglamorous chapters of this volume, doing their work.

EXECUTIVE LENS

A platform claim is audited per destination and phase, never accepted mission-wide.

MISSION LENS

The un-woken lander in rehearsal is the finding: survival reserves are not variables.

OPERATIONS LENS

Hazard scores in the wrong light decided a landing date. That is the system working.

WORKED EXAMPLE - AURORA ORBITAL, PART 2

LD5 granted in part: the licence that survived was smaller and worth more

Fictional worked example, continued. All figures illustrative.

Function	Claimed	Licensed at LD4
Cruise navigation and health (Wayfarer)	Rung 5, full autonomy	Rung 4 - survived, T3 envelope, demotion triggers live
Descent and landing (Selene)	Rung 4	Rung 4 - polar campaign passed, smaller signed ellipse
Surface traverses and science selection	Rung 4	Rung 3 - rehearsed, verified segments, charter audit live
Power and night survival	Rung 4	Rung 4 - after reserve restored as a signed constraint and wake rehearsed
ISRU production depended on for return	Rung 4	Refused - demonstration rate only; return planned on launched supplies
Learning in flight during descent	Standard practice	Refused - learning frozen in critical phases

Cost per verified mission day settled at 0.84 against a 0.98 pre-platform baseline, fully loaded, with contacts priced at scarcity. LD5 scale was granted in part: the platform extends to the next destination one envelope at a time, each earning its own licence through rehearsal and survival.

The vendor relationship improved under the new terms - envelopes, demotion triggers, charter audits - because the contract finally described the product Aurora actually needed: not full autonomy, but autonomy whose limits had been flown.

EXECUTIVE LENS

Scale in part is the normal, healthy outcome of an honest gate.

MISSION LENS

Refusing ISRU dependence kept the return plan honest - and funded.

OPERATIONS LENS

A smaller true licence compounds; a grand false one detonates far from home.

One page, to be read before any vehicle leaves Earth supervision

The Distance Standard. We license lunar and deep-space AI per destination, per vehicle, per light-time regime and per mission phase. Every decision that cannot wait for Earth is taken inside an envelope Earth signed in advance. Survival logic is deterministic. Learning is frozen in critical phases. Planetary protection outranks science, mass and schedule. Anyone can halt the autonomy before departure; in flight, envelopes expire and tiers demote automatically.

- We claim only the rung we survived: planned, simulated, rehearsed, survived, transferable.
- We design for the worst light-time and the longest silence, never the average contact.
- We never spend the survival reserve, and we rehearse the vehicle that does not wake.
- We audit what the vehicle chose, what it passed over and what it discarded.
- We count cost per verified mission day, with rehearsal, contacts and the retained anomaly team inside the number.

Print it. Sign it per mission and per destination. A standard that lives only in this playbook governs nothing.

EXECUTIVE LENS

Sign it per destination; an unsigned standard is a poster.

MISSION LENS

The standard is short because the contact is short.

OPERATIONS LENS

Hold the standard at departure gates, where it still costs only schedule.

SIX FAILURE PATTERNS

How lunar and deep-space programmes actually go wrong

Pattern	What happens	Antidote
1. The friendly rehearsal	Systems are proven in the wrong light, terrain or delay and licensed on it	Fidelity-gap list with every claim; failure-seeking campaigns per destination
2. The spent reserve	An optimiser quietly uses survival energy, margin or propellant to make a plan work	Signed central reserves no optimiser may spend
3. The confident position	Fused navigation hides a biased source and reports false precision	Source disagreement surfaced; uncertainty shipped with every solution
4. The silent drift	Science, triage or traverse taste drifts from the signed charter unnoticed	Pass-over and discard audits on a cadence
5. The heroic contact	The plan depends on one pass, relay or person that physics or rosters remove	Missed contact as a designed, rehearsed state
6. The un-woken vehicle	Hibernation or safe mode is stable for a weekend, not for the real silence	Safe states sized for the longest silence; no-wake case rehearsed

Every pattern was survivable in rehearsal and total at distance. The patterns repeat because each begins as a reasonable shortcut under schedule pressure, taken on Earth, paid for elsewhere.

EXECUTIVE LENS

Patterns are cheaper to recognise in this table than in a final telemetry frame.

MISSION LENS

Schedule pressure is the common ancestor; govern it, not just its children.

OPERATIONS LENS

Rehearsal at the real destination conditions is where patterns go to be caught cheaply.

30 QUESTIONS

Asked at every gate, answered with evidence or not at all

- Which destinations, vehicles, light-time regimes and phases does this licence cover?
- What rung is claimed, and what rehearsal or survival proves it?
- What is the worst light-time and longest silence, and is the plan designed to them?
- Which functions sit at each autonomy tier, and what demotes them automatically?
- Where does learning freeze, and who verified the freeze?
- What are the fidelity gaps in the rehearsal evidence?
- Is the landing / approach envelope signed, with divert and abort edges?
- Are survival reserves central, signed and unspendable by optimisers?
- Has the no-wake case been rehearsed end to end?
- Does every navigation solution ship uncertainty and source health?
- What happens, by design, on the missed contact?
- Are onboard discards and pass-overs audited, by whom, how often?
- What is the science charter, and when was model taste last audited against it?
- Which protection exclusions are hard-coded, and who owns them?
- Can the vehicle safe itself for the longest silence, in power and thermal?
- Are updates checker-verified, twin-flown and revertible without a healthy new image?
- What is the reporting contract, and what is the vehicle's self-report accuracy?
- Are approach corridors published and aborts passively safe?
- What ISRU output, if any, is depended upon, at what survived rung?
- Can crew isolate any habitat automation by hand, into a livable state?
- What are the tested environmental envelopes, and what demotes outside them?
- Are authorisation, protection category and partner terms closed at this gate?
- What is the fully loaded cost per verified mission day?
- Which savings depend on thinning rehearsal, contacts or the anomaly team?
- What is the DSN / relay contention position, in writing?
- Which components are shared across missions, and what is the blast radius?
- How are radiation events logged against anomalies?
- What evidence would make us reduce this licence tomorrow?
- Who can halt before departure, and when was a halt last honoured?
- What expires in flight, when, and what does expiry revert to?

EXECUTIVE LENS

Thirty questions, one standard: evidence or it did not happen.

MISSION LENS

Ask them at departure gates, not after the silence starts.

OPERATIONS LENS

Any unanswered question is a rung the licence has not earned.

Six gates, each with the evidence that opens it

Gate	Meaning	Evidence required
LD0 - Frame	Scope the distance mission	Destination map per phase; standing refusals signed; authorisation and protection category opened
LD1 - Plan	Envelopes and tiers designed	Autonomy tiers and envelopes drafted; science charter and reserves signed; reporting contract set
LD2 - Simulate	Claims tested in simulation	Dispersed campaigns with failure-seeking cases; twin at flight configuration; fidelity gaps listed
LD3 - Rehearse	Operations proven at real delay	Analogue and delay rehearsals passed with crews; no-wake, missed-contact and demotion cases flown on the ground
LD4 - License	Departure reliance at earned rungs	Per-phase licences; envelopes signed and expiring; halt honoured in drill
LD5 - Scale	Extend destinations and functions	Survival evidence per new destination; unit cost verified with rehearsal beside it

First 90 days

- **Days 1-30.** Build the destination map and envelope register; publish the standing refusals; sign the science charter and survival reserves.
- **Days 31-60.** Tier every function with demotion triggers; run the first failure-seeking simulation campaign; list every fidelity gap.
- **Days 61-90.** Rehearse at real light-time, including missed contact, night entry and a pre-departure halt; hold the first LD4 gate on evidence.

EXECUTIVE LENS

Gates open on rehearsals at real delay, not on simulations at zero delay.

MISSION LENS

The 90 days end with a shorter claim and a longer envelope register.

OPERATIONS LENS

LD5 adds destinations one envelope at a time; it never widens a licence by itself.

What an auditor, a regulator or an inquiry will eventually ask for

The evidence pack, per mission

- Envelope register history: every tier, limit, expiry and demotion, per phase.
- Rehearsal records with fidelity-gap lists and findings closure.
- Navigation scoring against withheld ground truth, per terrain and illumination.
- As-flown reconstructions: descents, traverses, hibernations and anomalies.
- Science charter audits: selections, pass-overs and onboard discards sampled.
- Planetary-protection records: bioburden, exclusion zones and containment verification.
- Command and update records with checker scores and rollback tests (Volume 6).
- Contact, relay and priority audits, including missed-contact behaviour.
- Halt, demotion and expiry log with signatures.

Sector annex - where the emphasis moves

Sector	Emphasis
Lunar surface missions	Polar illumination, lunar night survival, dust budgets, relay coverage
Mars and outer-system probes	Conjunction survival, longest-silence safe states, charter drift over years
Crewed lunar / Gateway	Crew override, life-support margins, passively safe rendezvous
Sample return	Containment determinism, protection provenance, ascent licensing
Commercial lunar delivery	Authorisation clarity, payload protection, honest rung claims to customers
ISRU demonstrators	Sustained rate at grade, feedstock-map verification, no premature dependence

EXECUTIVE LENS

Assemble the pack before anyone asks; afterwards is too late and it shows.

MISSION LENS

The annex moves emphasis, never the standard.

OPERATIONS LENS

The envelope register history is the evidence pack's spine.

EIGHT FILL-IN TEMPLATES

The licence, made paperwork

Template 1 - Distance Licence Card
Destination: ___ | Vehicle / phase: ___ | Rung: ___ | Light-time regime: ___ | Expiry: ___ | Authority: ___

Template 2 - Autonomy Envelope Record
Function: ___ | Tier: ___ | Hard limits: ___ | Abort conditions: ___ | Demotion triggers: ___ | Signed: ___

Template 3 - Landing / Approach Envelope
Ellipse / corridor: ___ | Hazard thresholds: ___ | Reserves: ___ | Divert limits: ___ | Wave-off conditions: ___

Template 4 - Survival Budget Sheet
Period / night: ___ | Reserve energy: ___ | Heater minimums: ___ | Wake margins: ___ | Verified by rehearsal: ___

Template 5 - Missed-Contact Plan
Silence window: ___ | Permitted actions: ___ | Stored / dropped data: ___ | Exit checks: ___ | Owner: ___

Template 6 - Science Charter Audit
Period: ___ | Chosen: ___ | Passed over: ___ | Outside charter: ___ | Discards sampled: ___ | Findings: ___

Template 7 - Rehearsal and Fidelity-Gap Record
Scenario: ___ | Delay injected: ___ | Failures injected: ___ | Fidelity gaps: ___ | Findings / owners: ___

Template 8 - Verified Mission Day Workbook
Period: ___ | Fully loaded cost: ___ | Verified days (survival / science): ___ | Cost per verified day: ___ | Claimed / released / banked: ___

Envelopes, budgets, silences, charters, rehearsals and cost - each short enough to complete before a contact window closes, which is when several of them will be used. Paperwork from logs is evidence; from memory, fiction.

The volume in twenty lines, and the series in ten

- License per destination, vehicle, light-time regime and phase.
- A simulated landing is not a landed mission.
- Decisions that cannot wait stay inside envelopes Earth signed.
- Autonomy demotes automatically; promotion is earned at a gate.
- Learning is frozen in critical phases.
- No optimiser spends the survival reserve.
- Design for the worst light-time and the longest silence.
- Navigation ships its uncertainty, or it does not ship.
- In descent, AI chooses only among futures Earth accepted.
- Count verified metres and days, not odometer claims.
- Missed contacts are a designed state, rehearsed.
- Audit selections, pass-overs and discards alike.
- Planetary protection outranks science, mass and schedule.
- Survival FDIR is deterministic, independent and simple.
- Never activate what you cannot revert, on the edge of a silence.
- Rehearsal claims carry their fidelity gaps.
- Self-report accuracy is the trust metric at distance.
- Depend only on ISRU already produced, sustained, at grade.
- Cost per verified mission day, with rehearsal and contacts inside.
- Anyone can halt before departure. A smaller true licence beats a grand false one.

AI in Space and Orbital Operations - the series

Vol	Title	Gates	Status
1	Space Strategy	SS0-SS5	Built
2	Mission Design and Autonomy	MD0-MD5	Built
3	Satellite and Constellation Operations	CO0-CO5	Built
4	Earth Observation and Data	EO0-EO5	Built
5	Conjunction, Debris and Orbital Safety	CD0-CD5	Built
6	Ground Systems and Mission Software	GS0-GS5	Built
7	Lunar and Deep-Space Operations	LD0-LD5	Built - this volume
8	Space Security and Resilience	SR0-SR5	Next
9	Space Economics and Fleet Finance	EF0-EF5	Planned
10	Space Capstone and Assurance	SC0-SC5	Planned

Volume 8, Space Security and Resilience, is next. The series closes at ten with the Space Capstone and Assurance.